

Tourism, Takeoff and Growth: A Quantitative Analysis of Macau

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Abstract

This study develops an open-economy Schumpeterian growth model that features endogenous takeoff and a tourism sector. Using this model, we study the role of tourism in shaping the transition of an economy from stagnation to growth. Suppose leisure preference is strong. Then, an expansion in tourism triggers an earlier takeoff and leads to higher transitional growth when tourism reliance is low, but delays takeoff and leads to lower transitional growth when tourism reliance is high. We also calibrate the model to data in Macau. Under our calibrated values, the quantitative results suggest that Macau's transitional growth is maximized when its tourism share is about 60% of the economy, and the rapid expansion in tourism after Macau's return to China triggers its transition to the era of quality-driven growth five years earlier. To provide empirical evidence for our theoretical prediction, we use cross-country panel data and find that the relationship between tourism and TFP growth follows an inverted-U pattern. In addition, we construct Bartik-style measures and the inverted-U pattern between tourism and TFP growth remains robust.

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“Tourism growth can and should lead to economic prosperity, jobs and resources to fund environmental protection and cultural preservation.” UN Tourism (2017)¹

1 Introduction

Macau, a world-renowned tourism hub and one of the highest per-capita-GDP economies in the world, has experienced rapid economic growth fueled by a booming tourism sector. However, despite its initial economic success, Macau has struggled to transition beyond tourism-dependent industries and develop high-tech industries, raising concerns about the long-term sustainability of its growth.² These contrasting experiences highlight the dual role of tourism in economic development and naturally raise an interesting question: does tourism contribute to the transition of the economy from stagnation to sustained innovation-driven growth? To address this question, we develop a small-open-economy version of the Schumpeterian growth model with a tourism sector and endogenous takeoff. Our key findings are as follows. Suppose leisure preference is weak. Then, an expansion in tourism delays takeoff and reduces the post-takeoff transitional growth rate. Suppose leisure preference is strong. Then, an expansion in tourism has non-monotonic effects on both takeoff and transitional growth: the effect is positive when the economy does not heavily rely on tourism, but this effect becomes negative when tourism reliance is high. However, tourism has no effect on the long-run growth rate.

The mechanism underlying the above results is that an expansion in tourism affects the economy by changing firm size. A larger (smaller) firm size makes innovation more (less) profitable, thereby leading to an earlier (a delayed) takeoff from stagnation to innovation-driven growth, as well as higher (lower) post-takeoff transitional growth. In the short run, a higher share of labor allocated to production is associated with a larger firm size. Therefore, the short-run effects of tourism on the timing of takeoff and economic growth depend on how tourism affects the production labor share. While a recent study by Chu *et al.* (2024) shows how tourism affects labor allocation and economic growth in the modern era, it does not consider takeoff from stagnation to sustained growth. Our study complements and extends Chu *et al.* (2024) by introducing endogenous takeoff and examining how tourism shapes the entire phase-transition dynamics of the economy. In addition, we calibrate the model to data in Macau and quantify the effects of tourism on its transitional dynamics. This broader perspective is crucial for understanding the implications of tourism not only on growth, but also on the transition of the economy from stagnation to growth.

Specifically, since foreign tourists not only consume the domestic final good but also local tourism services, an expansion in tourism affects the production labor share through two opposing channels. On the one hand, it increases demand for the domestic final good, which requires more labor for production and thereby raises the level of employment. We call this positive effect the *employment effect*. On the other hand, it also increases demand for tourism services, which may shift labor away from production toward tourism services. We call this

¹<https://www.unwto.org/archive/global/press-release/2017-08-15/tourism-growth-not-enemy-it-s-how-we-manage-it-counts>

²Other tourism-dependent economies, such as the Maldives, the Bahamas, and Fiji have also faced challenges similar to those of Macau. In contrast, some economies, such as Spain and Dubai, have benefited from booming tourism and gradually developed high-tech industries, notably in renewable energy and advanced manufacturing.

negative effect the *reallocation effect*.³ Which effect dominates is governed by the strength of leisure preference and the economy’s reliance on tourism.

When leisure preference is weak, households already devote most of their available time to work. As a result, an expansion in tourism cannot significantly increase labor supply. Consequently, the employment effect is small and the reallocation effect always dominates. Therefore, under weak leisure preference, an expansion in tourism delays takeoff and reduces the post-takeoff transitional growth rate. When leisure preference is strong, whether the employment or reallocation effect dominates depends on how much the economy relies on tourism. If the economy has a low reliance on tourism, job opportunities in the tourism sector are relatively limited, so the reallocation effect is small and the employment effect dominates. However, if the economy is highly reliant on tourism, most of the additional labor flows into the tourism sector, and the reallocation effect dominates. As a result, under strong leisure preference, the effects of tourism expansion on takeoff and transitional growth are non-monotonic and follow an inverted-U pattern with respect to the transitional growth rate. In the long run, however, endogenous market structure governs firm entry, and thus eventually leads to a steady-state level of firm size that is independent of the production labor share. Consequently, tourism has no effect on the long-run growth rate.

Our quantitative analysis focuses on Macau to gain insights into how tourism shapes the transition of an economy that is heavily reliant on tourism. In the quantitative analysis, we calibrate our model to Macau’s data and quantitatively validate the inverted-U relationship between its tourism expansion and technological growth. Specifically, Macau’s transitional growth is maximized when its tourism share is about 60% of the economy. In the data, tourism-related industries (including gambling and non-gambling activities) account for a substantial share of Macau’s gross value added, reaching about 70% in some peak years and slightly below 60% in recent years. Moreover, the rapid expansion in tourism after Macau’s return to China in 1999 triggers its transition to the era of quality-driven growth five years earlier.

In our empirical analysis, we rely on cross-country panel data to provide broader empirical implications. Specifically, we use cross-country panel data with total factor productivity (TFP) as a proxy for innovation-driven growth. Unlike most empirical studies that use GDP or GDP per capita as the proxy, which may mechanically reflect tourism revenues, our proxy captures how tourism affects innovation. Our empirical results imply that the growth effect of tourism weakens and eventually becomes negative as tourism reliance increases, consistent with our theoretical predictions. In addition, we construct Bartik-style tourism measures, and the inverted-U relationship between tourism and TFP growth remains robust.

This study builds on the literature on innovation-driven growth. Romer (1990) develops the first R&D-based growth model that emphasizes the role of variety-expanding innovation in driving economic growth. Early studies by Aghion and Howitt (1992), Grossman and Helpman (1991) and Segerstrom *et al.* (1990) instead highlight the role of quality-improving innovation by developing quality-ladder Schumpeterian growth models. Subsequent studies incorporate both variety expansion and quality improvement into the Schumpeterian growth model by adopting

³This reallocation effect marks a key difference between the mechanism of tourism in this study and that of exports in Chu *et al.* (2023), where only the employment effect is present.

an endogenous market structure.⁴ In this vein, Smulders and van de Klundert (1995) and Peretto (1998, 1999) develop variants with creative accumulation, while Howitt (1999) presents a variant with creative destruction. Recent empirical evidence from Garcia-Macia *et al.* (2019) indicates that innovation is mostly introduced by incumbents rather than entrants, consistent with the notion of creative accumulation. Our study contributes to this literature by developing an open-economy Schumpeterian growth model with a tourism sector and endogenous takeoff to explore how tourism affects the entire dynamics of the economy from stagnation to growth.

This study also relates to the literature on tourism and economic growth. An early study by Sinclair (1998) highlights the dual role of tourism in economic development.⁵ Subsequent studies by Pina and Martinez-Garcia (2013) and Liu and Wu (2019) develop capital-based growth models to explore how tourism affects capital accumulation and economic growth. Some recent studies incorporate a tourism sector into R&D-based growth models; see Albaladejo and Martinez-Garcia (2015) and Hamaguchi (2020) for variants based on Romer (1990), and Barrera and Garrido (2018) for a variant based on Aghion and Howitt (1992). These models consider only one dimension of innovation, focusing on growth driven by either variety expansion or quality improvement. Our study is most closely related to Chu *et al.* (2024), who consider two dimensions of innovation and examine the effects of tourism on innovation-driven growth in the modern era. However, they do not consider endogenous takeoff from stagnation to sustained growth. Compared to Chu *et al.* (2024), our study develops a more general model that incorporates the endogenous takeoff of the economy. In addition, our model considers exports and the social return to variety, which allows us to perform an empirically relevant quantitative analysis for Macau.⁶ We contribute to this literature by developing a small-open-economy version of the Schumpeterian growth model with a tourism sector and endogenous takeoff. This model allows us to examine how tourism affects the entire phase-transition of the economy.

Finally, this study draws on insights from the literature on endogenous takeoff from stagnation to sustained economic growth. Galor and Weil (2000) provide the seminal study in unified growth theory (UGT), which emphasizes the endogenous transition of the economy from stagnation to modern economic growth. Subsequent studies explore the effects of various prehistorical and historical factors on this transition and provide empirical support for UGT; see for example, Galor and Moav (2002), Galor and Mountford (2008), Galor *et al.* (2009) and Ashraf and Galor (2011). In addition, Galor (2005, 2011) provides comprehensive reviews of UGT. While UGT provides valuable insights into endogenous takeoff, it typically does not explicitly model the innovation-driven growth process. Building on the Schumpeterian perspective, Peretto (2015) develops a closed-economy R&D-based model that features endogenous takeoff from pre-industrial stagnation to modern innovation-driven growth. A recent study by Chu *et al.* (2023) extends the Peretto model to a small-open-economy version to examine the effects of exports, while their model does not incorporate a tourism sector.⁷ Our study contributes

⁴Empirical evidence from Laincz and Peretto (2006), Ha and Howitt (2007), Madsen (2008), and Ang and Madsen (2011) supports second-generation Schumpeterian growth models that feature both dimensions of innovation.

⁵See also Copeland (1991) and Chao *et al.* (2006) for a discussion on de-industrialization effects of tourism.

⁶Our study also enriches the empirical analysis in Chu *et al.* (2024) by using a larger sample of cross-country panel data, adopting alternative proxies, and constructing plausibly exogenous tourism exposure measures.

⁷See related studies by Iacopetta and Peretto (2021), Chu *et al.* (2020a), Chu *et al.* (2020b), and Chu *et al.* (2022), which also contribute to this strand of literature.

to this literature by introducing a tourism sector to an open-economy Schumpeterian growth model with takeoff, which allows us to explore how tourism affects endogenous takeoff and innovation-driven growth.

The rest of this study is organized as follows. Section 2 develops an open-economy version of the Schumpeterian model with a tourism sector and endogenous takeoff. Section 3 shows the theoretical results. Section 4 performs our quantitative analysis focusing on Macau. Section 5 presents supportive empirical evidence for our theoretical findings, and Section 6 concludes.

2 Model

We build on the model in Peretto (2015), which features takeoff from stagnation to innovation-driven growth. Furthermore, Chu *et al.* (2023) introduce international trade to the Peretto model to explore how exports affect endogenous takeoff and economic growth. This study differs from the above studies by incorporating a tourism sector to examine how tourism shapes the entire dynamics of the economy.

While Chu *et al.* (2024) also introduce a tourism sector to the Schumpeterian growth model, they focus only on economic growth in the modern era and do not consider endogenous takeoff. Instead, our model captures the entire transition of the economy from stagnation to growth and further incorporates exports and the social return to variety, which enable us to conduct an empirically relevant quantitative analysis for Macau.

2.1 Household

The representative household's preferences are described by the following utility function

$$U = \int_0^\infty e^{-(\rho-\lambda)t} \left[\ln c_t + \delta \ln(1 - l_t) + \psi \frac{(\iota_t)^{1-\epsilon}}{1-\epsilon} \right] dt, \quad (1)$$

where c_t and ι_t denote per capita consumption of a domestic final good and an imported good, respectively. The parameter $\delta > 0$ measures the household's preference for leisure $1 - l_t$, where l_t is per capita labor supply. In addition, $\rho > 0$ is the subjective discount rate, while $\lambda \in (0, \rho)$ is the population growth rate. $\psi > 0$ and $\epsilon \in (0, 1)$ are, respectively, the preference parameter and the inverse of the intertemporal elasticity of substitution for the imported good. The asset-accumulation equation is

$$\dot{a}_t = (r_t - \lambda)a_t + w_t l_t - c_t - p_t \iota_t, \quad (2)$$

where we choose the domestic final good as the numeraire. w_t denotes the real wage rate. r_t denotes the real interest rate and a_t denotes per capita asset value. Finally, p_t is the price of the imported good.

The Euler equation characterizes the optimal consumption path

$$\frac{\dot{c}_t}{c_t} = r_t - \rho. \quad (3)$$

Dynamic optimization yields the supply of labor

$$l_t = 1 - \frac{\delta c_t}{w_t}, \quad (4)$$

and the conditional demand function for the imported good

$$\iota_t = \left(\frac{\psi c_t}{p_t} \right)^{\frac{1}{\epsilon}}. \quad (5)$$

2.2 Domestic Final Good

Domestic competitive firms produce the final good:

$$Y_t = \int_0^{N_t} X_t^\theta(i) \left[Z_t^\alpha(i) Z_t^{1-\alpha} \frac{L_{y,t}}{N_t^{1-\sigma}} \right]^{1-\theta} di, \quad (6)$$

where Y_t is the final output, and N_t measures the number of intermediate-good varieties. For each intermediate good i , $X_t(i)$ and $Z_t(i)$ denote its quantity and quality, respectively. The average quality is defined as $Z_t \equiv \int_0^{N_t} Z_t(i) di / N_t$. In addition, $1 - \theta$ measures labor intensity, and $1 - \alpha$ governs technology spillovers, where $\{\theta, \alpha\} \in (0, 1)$. $L_{y,t}$ is production labor input. The parameter $\sigma \in (0, 1)$ is the social return to variety, and $1 - \sigma$ captures the congestion effect that removes the strong scale effect in our model.

Profit maximization implies the following demand function for intermediate good i

$$X_t(i) = \left[\frac{\theta}{P_t(i)} \right]^{\frac{1}{1-\theta}} Z_t^\alpha(i) Z_t^{1-\alpha} \frac{L_{y,t}}{N_t^{1-\sigma}}. \quad (7)$$

$P_t(i)$ denotes the price of intermediate good i . Under the zero-profit condition, payments to production labor and intermediate inputs satisfy

$$(1 - \theta)Y_t = w_t L_{y,t}, \quad (8)$$

$$\theta Y_t = \int_0^{N_t} P_t(i) X_t(i) di. \quad (9)$$

2.3 Intermediate Goods and In-house R&D

Each monopolistic firm $i \in [0, N_t]$ produces its intermediate good i under a linear one-to-one technology. Therefore, producing $X_t(i)$ units of intermediate good i requires $X_t(i)$ units of the final good as input. Firm i also incurs an operating cost of $\phi Z_t^\alpha(i) Z_t^{1-\alpha}$ units of the final good, where the operating cost parameter $\phi > 0$. Before R&D, monopolistic firm i earns profit given by

$$\Pi_t(i) = P_t(i) X_t(i) - X_t(i) - \phi Z_t^\alpha(i) Z_t^{1-\alpha}. \quad (10)$$

A higher quality level raises the demand for intermediate good i as shown in (7). To improve the quality of its product, firm i invests $R_t(i)$ units of the final good to conduct in-house R&D:

$$\dot{Z}_t(i) = R_t(i). \quad (11)$$

Monopolistic firm i 's value is expressed as

$$V_t(i) = \int_t^\infty \exp \left(- \int_t^s r_u du \right) [\Pi_s(i) - R_s(i)] ds, \quad (12)$$

where $\Pi_s(i) - R_s(i)$ denotes the net profit after R&D. Firm i maximizes its value in (12) subject to (7), (10) and (11). Dynamic optimization implies that each monopolistic firm i sets its price as

$$P_t(i) = \min \left\{ \mu, \frac{1}{\theta} \right\} = \mu, \quad (13)$$

where $\mu \in (1, 1/\theta)$ denotes the marginal cost of producing intermediate good i with quality level $Z_t(i)$ for competitive firms. To drive these competitive firms out of the market, the prices of intermediate goods are set to μ . Following previous studies, we assume a common initial quality level across intermediate goods, i.e., $Z_0(i) = Z_0$ for $i \in [0, N_t]$. This assumption leads to a symmetric equilibrium; namely, $Z_t(i) = Z_t$, $X_t(i) = X_t$ and $\Pi_t(i) = \Pi_t$ for $i \in [0, N_t]$. Under the symmetric equilibrium, the dynamic optimization of monopolistic firms yields the rate of return to quality-improving innovation

$$r_t^q = \alpha \left[(\mu - 1) \frac{X_t}{Z_t} - \phi \right], \quad (14)$$

which positively depends on quality-adjusted firm size, X_t/Z_t . Using (7), (13) and the symmetric condition $Z_t(i) = Z_t$ yields

$$\frac{X_t}{Z_t} = \left(\frac{\theta}{\mu} \right)^{\frac{1}{1-\theta}} \frac{L_{y,t}}{N_t^{1-\sigma}} = \left(\frac{\theta}{\mu} \right)^{\frac{1}{1-\theta}} \frac{L_t l_{y,t}}{N_t^{1-\sigma}}. \quad (15)$$

$l_{y,t}$ is the share of total labor L_t allocated to production. For simplicity, the state-variable component of the quality-adjusted firm size is defined as

$$x_t \equiv \left(\frac{\theta}{\mu} \right)^{\frac{1}{1-\theta}} \frac{L_t}{N_t^{1-\sigma}}. \quad (16)$$

In this notation, $x_t l_{y,t} = X_t/Z_t$ denotes the quality-adjusted firm size. Hereafter, we refer to quality-adjusted firm size simply as *firm size* when no confusion arises. Then, we re-express (14) as

$$r_t^q = \alpha [(\mu - 1)x_t l_{y,t} - \phi]. \quad (17)$$

2.4 Entrants

To seek positive monopolistic profits, new entrants supply differentiated intermediate goods, each characterized by the average quality level of Z_t .⁸ Firms incur a sunk entry cost proportional to X_t , scaled by the entry-cost parameter $\beta > 0$.⁹ The free-entry condition takes the form

$$V_t = \beta X_t. \quad (18)$$

From the asset-pricing equation, the rate of return is determined as:

$$r_t = \frac{\Pi_t - R_t}{V_t} + \frac{\dot{V}_t}{V_t}. \quad (19)$$

⁸Following Peretto (2015), we adopt this assumption to keep the model tractable.

⁹One can think of this as households financing the sunk entry cost to acquire equity in the new firm.

We substitute (10), (11), (13), (15), (16) and (18) into (19) to obtain the rate of return to variety-expanding innovation

$$r_t^e = \frac{1}{\beta} \left(\mu - 1 - \frac{\phi + z_t}{x_t l_{y,t}} \right) + \frac{\dot{l}_{y,t}}{l_{y,t}} + \frac{\dot{x}_t}{x_t} + z_t, \quad (20)$$

where $z_t \equiv \dot{Z}_t/Z_t$ denotes quality growth. Consistent with the rate of return to quality-improving innovation in (17), the rate of return to variety-expanding innovation in (20) also positively depends on firm size, $x_t l_{y,t}$.

2.5 Tourism and International Trade

Foreign tourists consume not only the domestic final good but also tourism services. To satisfy the demand for tourism services, $L_{s,t}$ units of labor are employed in the tourism sector.¹⁰ Our model considers a small open economy where both foreign tourism demand and export demand are exogenous. In our setting, a constant share $\chi > 0$ of the final good is exported and a constant share $\tau > 0$ is consumed by foreign tourists. The parameter τ measures tourism demand; hereafter, we simply refer to τ as tourism. Labor employed in the tourism sector follows $L_{s,t} = \xi \tau l_t L_t$, where $\xi \in (0, 1/\tau)$.¹¹ Intuitively, an expansion in tourism τ also increases labor demand for tourism services.¹² The economy uses revenue from tourism and exports to pay for its imports. The balanced-trade condition is

$$p_t l_t L_t = \tau Y_t + w_t L_{s,t} + \chi Y_t. \quad (21)$$

2.6 Equilibrium

The equilibrium is defined as a collection of time paths of allocations $\{a_t, \iota_t, c_t, Y_t, L_{y,t}, L_{s,t}, l_t, l_{y,t}, l_{s,t}, X_t(i), R_t(i)\}$ and prices $\{r_t, w_t, p_t, P_t(i), V_t(i)\}$ that satisfy the following conditions:

- the representative household chooses $\{c_t, l_t, \iota_t\}$ to maximize its lifetime utility, taking $\{r_t, w_t, p_t\}$ as given;
- competitive final-good firms choose factor inputs $\{L_{y,t}, X_t(i)\}$ to maximize profit, taking $\{w_t, P_t(i)\}$ as given;
- for each $i \in [0, N_t]$, monopolistic intermediate-good firm i chooses price and R&D investment $\{P_t(i), R_t(i)\}$ to maximize firm value $V_t(i)$, taking r_t as given;
- new entrants make entry decisions, taking V_t as given;

¹⁰The requirement of labor for providing tourism services is the key characteristic that distinguishes the mechanism of tourism expansion from the export-led mechanism in Chu *et al.* (2023).

¹¹Chu *et al.* (2024) simply set $\sigma = 0$ and $\xi = 1$ in their model. We generalize these parameters in order to conduct an empirically relevant quantitative analysis for Macau.

¹²Our reduced-form specification can be micro-founded. Suppose tourism services are produced under perfect competition with production function $s_t = L_{s,t}/\xi$, where ξ is the inverse of productivity. The zero-profit condition implies that the price of tourism services is $p_{s,t} = \xi w_t$. Also, the demand for tourism services is $D_{s,t} = \tau l_t L_t$, which is proportional to labor force $l_t L_t$ to ensure balanced growth. The market-clearing condition $s_t = D_{s,t}$ implies $L_{s,t} = \xi \tau l_t L_t$.

- the aggregate value of monopolistic firms is equal to the total value of household assets, such that $N_t V_t = a_t L_t$;
- the balanced-trade condition is satisfied, such that $p_t \iota_t L_t = \tau Y_t + w_t L_{s,t} + \chi Y_t$;
- the final-good market clears, $Y_t = c_t L_t + N_t(X_t + R_t + \phi Z_t) + \dot{N}_t \beta X_t + \tau Y_t + \chi Y_t$; and
- the labor market clears, with total labor allocated across production and tourism services $L_t l_t = L_{y,t} + L_{s,t} = L_t l_{y,t} + L_t l_{s,t}$.

2.7 Aggregation

Combining (6), (7) and (13), and imposing symmetry $Z_t = Z_t(i)$ for $i \in [0, N_t]$, we derive the aggregation production function for the final good in equilibrium:

$$Y_t = \left(\frac{\theta}{\mu}\right)^{\frac{\theta}{1-\theta}} N_t^\sigma Z_t L_{y,t} = \left(\frac{\theta}{\mu}\right)^{\frac{\theta}{1-\theta}} N_t^\sigma Z_t L_t l_{y,t}, \quad (22)$$

where the second equality uses $L_{y,t} = L_t l_{y,t}$. Defining per capita domestic final output as $y_t \equiv Y_t/L_t$, its growth rate is

$$g_t \equiv \frac{\dot{y}_t}{y_t} = \sigma n_t + z_t + \frac{\dot{l}_{y,t}}{l_{y,t}}, \quad (23)$$

where $n_t \equiv \dot{N}_t/N_t$ and $z_t \equiv \dot{Z}_t/Z_t$, respectively, denote the growth rates of variety and quality.

2.8 Dynamics

According to (17) and (20), a larger firm size leads to a higher rate of return to innovation. This implies that innovation becomes more profitable as firm size increases. We consider a realistic sequence in which variety-expanding innovation comes before quality-improving innovation, which is consistent with the pattern observed in advanced economies. Our model features two threshold values of firm size that activate variety-expanding and quality-improving innovation, respectively.¹³ When firm size x_t reaches the first threshold x_N , variety expansion becomes profitable and begins to take place. Then, when it reaches the second threshold x_Z , quality improvement becomes profitable and begins to take place.¹⁴ We impose the following parameter condition to ensure stability of our model:

$$\beta\phi > \frac{1}{\alpha} \left[\mu - 1 - \beta \left(\rho + \frac{\sigma\lambda}{1-\sigma} \right) \right] > \mu - 1. \quad (24)$$

According to (16), the state variable x_t evolves as follows:

$$\frac{\dot{x}_t}{x_t} = \lambda - (1 - \sigma)n_t. \quad (25)$$

¹³See the derivation in Section 3.

¹⁴In Appendix B, we derive a parameter condition that ensures this more realistic sequence, in which variety expansion happens before quality improvement.

As we will show in Section 3, variety growth n_t is a monotonically increasing function of x_t . Therefore, (25) characterizes a stable dynamic process, and x_t eventually converges to its steady-state level, denoted as x^* .

Given the parameter condition specified in (24), the economy evolves through three phases: (1) In the pre-industrial era, the state variable $x_t \leq x_N$ and firm size remains below the threshold for variety expansion. At this early stage, innovation is not profitable, so neither variety expansion nor quality improvement occurs.¹⁵ (2) In the first phase of the industrial era, the state variable $x_N < x_t \leq x_Z$, firm size exceeds the first threshold, making variety expansion profitable. In this phase, variety-expanding innovation takes place, but quality-improving innovation remains absent because firm size has not yet reached the second threshold. (3) In the second industrial phase, the state variable x_t exceeds the second threshold x_Z . In this phase, firm size is sufficiently large to support both variety expansion and quality improvement. In the long run, the economy approaches a balanced growth path as the state variable x_t converges to its steady-state value x^* .

Before analyzing how tourism expansion affects the entire transition of the economy, we present the steady-state properties of the consumption-output ratio and production labor share in Lemma 1.

Lemma 1 (*Consumption-output ratio and production labor share*) Assume that $1 - \theta - \tau - \chi > 0$. At any time t , both the consumption-output ratio c_t/y_t and production labor share $l_{y,t}$ adjust instantaneously to their unique steady-state values. The steady-state expressions for the consumption-output ratio and production labor share are given by:

$$\frac{c_t}{y_t} = \left(\frac{c_t}{y_t} \right)^* = \begin{cases} 1 - \theta - \tau - \chi & 0 \leq x_t \leq x_N \\ \frac{(\rho - \lambda)\beta\theta}{\mu} + 1 - \theta - \tau - \chi & x_N < x_t < \infty \end{cases}; \quad (26)$$

$$l_{y,t} = l_y^* = \begin{cases} \left[\frac{1}{1 - \xi\tau} + \frac{\delta}{1 - \theta} (1 - \theta - \tau - \chi) \right]^{-1} & 0 \leq x_t \leq x_N \\ \left\{ \frac{1}{1 - \xi\tau} + \frac{\delta}{1 - \theta} \left[\frac{(\rho - \lambda)\beta\theta}{\mu} + 1 - \theta - \tau - \chi \right] \right\}^{-1} & x_N < x_t < \infty \end{cases}. \quad (27)$$

Proof. See Appendix B. ■

By differentiating the production labor share in (27) with respect to tourism τ , we obtain

$$\frac{\partial l_y^*}{\partial \tau} = (l_y^*)^2 \left[\frac{\delta}{1 - \theta} - \frac{\xi}{(1 - \xi\tau)^2} \right]. \quad (28)$$

If and only if the following condition holds, tourism expansion raises the production labor share

$$\tau < \frac{1}{\xi} \left(1 - \sqrt{\xi \frac{1 - \theta}{\delta}} \right). \quad (29)$$

If $\delta < \xi(1 - \theta)$, (29) never holds, so that an expansion in tourism reduces the production labor share. If $\delta > \xi(1 - \theta)$, the effect of tourism expansion on the production labor share

¹⁵We assume that intermediate goods are produced by competitive firms with a constant marginal cost of μ before the industrial era. In Appendix C, we also explore an alternative scenario where monopolistic firms engage in production in the pre-industrial era; however, the dynamics become less realistic.

follows an inverted-U pattern. Specifically, the effect of an expansion in tourism is positive when $\xi\tau < 1 - \sqrt{\xi(1-\theta)/\delta}$, but becomes negative when $\xi\tau > 1 - \sqrt{\xi(1-\theta)/\delta}$. The economic intuition is as follows. An expansion in tourism increases the demand for both the domestic final good and local tourism services, which reflects two opposing effects: the employment effect and the reallocation effect. On the one hand, the expansion of market size requires more labor, which increases the overall employment. On the other hand, the higher demand for tourism services may attract labor away from production to the tourism sector. When leisure preference is weak, households already devote most of their available time to work. In this case, an expansion in tourism cannot significantly increase labor supply. As a result, the employment effect is small and the reallocation effect always dominates. Therefore, under weak leisure preference, an expansion in tourism reduces the production labor share. When leisure preference is strong, the reliance on tourism determines which effect dominates. Specifically, if the economy has low reliance on tourism, job opportunities in the tourism sector are relatively limited, so the reallocation effect is small and the employment effect dominates. In contrast, if the economy is highly reliant on tourism, most of the additional labor is drawn into the tourism sector, and the reallocation effect dominates. Therefore, under strong leisure preference, an expansion in tourism increases the production labor share when the economy has low tourism reliance, while it decreases the production labor share when tourism reliance is high.

3 Tourism and Takeoff

This section explores how tourism shapes the dynamic transition of the economy from stagnation in the pre-industrial era to innovation-driven growth in the industrial era.

3.1 The Pre-Industrial Era

In the pre-industrial era, neither variety expansion nor quality improvement occurs because firm size is too small to make innovation profitable, i.e., $n_t = z_t = 0$. Lemma 1 suggests that production labor share remains constant over time, so that $\dot{l}_{y,t}/l_{y,t} = 0$. Using (23), per capita final output growth is given by

$$g_t = \sigma n_t + z_t + \frac{\dot{l}_{y,t}}{l_{y,t}} = 0. \quad (30)$$

Imposing $n_t = 0$ in (25) gives the evolution of the state variable x_t

$$\frac{\dot{x}_t}{x_t} = \lambda - (1 - \sigma)n_t = \lambda. \quad (31)$$

Therefore, x_t grows at a constant rate $\lambda > 0$ in this era. As a result, firm size $x_t l_y^*$ would exceed the threshold for variety-expanding innovation within finite time, thereby leading the economy to enter the first industrial phase.

3.2 The First Industrial Phase

In the first industrial phase, the firm size exceeds the first threshold but remains below the second threshold, so variety expansion is active (i.e., $n_t > 0$) but quality improvement is absent

(i.e., $z_t = 0$). Using (3), (20), (23), $z_t = 0$, $g_t \equiv \dot{y}_t/y_t = \dot{c}_t/c_t$ and $\dot{l}_{y,t}/l_{y,t} = 0$ yields variety growth as

$$n_t = \frac{1}{\beta} \left(\mu - 1 - \frac{\phi}{x_t l_y^*} \right) + \lambda - \rho, \quad (32)$$

where we impose $r_t = r_t^e$. The growth rate of variety is positive if and only if

$$x_t l_y^* > \frac{\phi}{\mu - 1 - \beta(\rho - \lambda)}, \quad (33)$$

which implies that the economy enters the first industrial phase when the firm size $x_t l_y^*$ exceeds the first threshold $\phi/[\mu - 1 - \beta(\rho - \lambda)]$. We rearrange (33) and obtain

$$x_t > x_N \equiv \frac{\phi}{\mu - 1 - \beta(\rho - \lambda)} \frac{1}{l_y^*}. \quad (34)$$

This suggests that x_N is decreasing in production labor share l_y^* . Let x_0 denote the initial value of the state variable x_t . Since x_t grows at the constant rate λ in the pre-industrial era, the timing of takeoff is given by $T_N = (\ln x_N - \ln x_0)/\lambda$. A higher production labor share lowers x_N , leading to a smaller value of T_N and thus an earlier takeoff. More explicitly, although tourism does not affect the dynamics of the state variable x_t , it affects the production labor share l_y^* and thus the firm size $x_t l_y^*$. Therefore, according to (33), tourism shapes the timing of takeoff through its effect on the firm size.

We combine (23) and (32) to obtain the transitional growth rate during this phase

$$g_t = \sigma n_t = \frac{\sigma}{\beta} \left(\mu - 1 - \frac{\phi}{x_t l_y^*} \right) - \sigma(\rho - \lambda), \quad (35)$$

which is increasing in production labor share l_y^* . Given the comparative statics of the production labor share with respect to tourism τ , Proposition 1 summarizes the effects of tourism in the first industrial phase.

Proposition 1 *(The effect of an expansion in tourism in the first industrial phase) With $\delta < \xi(1 - \theta)$, leisure preference is weak; in this case, an expansion in tourism delays takeoff from stagnation to innovation-driven growth and lowers its transitional growth. With $\delta > \xi(1 - \theta)$, leisure preference is strong; in this case, the effects of tourism expansion on takeoff and transitional growth are non-monotonic: it leads to an earlier (a delayed) takeoff and raises (lowers) the transitional growth rate when tourism reliance is low (high).*

Proof. Recall that the timing of takeoff is given by $T_N = (\ln x_N - \ln x_0)/\lambda$ and the transitional growth rate in this phase is given by (35). Therefore, a higher production labor share lowers x_N , leading to an earlier takeoff, and raises transitional growth. Combining these results with Lemma 1 proves Proposition 1. ■

In this phase, (25) and (32) characterize the dynamics of the state variable x_t

$$\frac{\dot{x}_t}{x_t} = \lambda - (1 - \sigma) \left[\frac{1}{\beta} \left(\mu - 1 - \frac{\phi}{x_t l_y^*} \right) + \lambda - \rho \right]. \quad (36)$$

Given the parameter condition (24), the state variable x_t keeps growing and exceeds x_Z within finite time. At that point, firm size $x_t l_y^*$ crosses the threshold for quality improvement, and the economy transitions into the second industrial phase.

3.3 The Second Industrial Phase

Both variety expansion and quality improvement are active (i.e., $n_t > 0$ and $z_t > 0$) in the second industrial phase. Combining (3), (17), (20), (23), $g_t \equiv \dot{y}_t/y_t = \dot{c}_t/c_t$ and $\dot{l}_{y,t}/l_{y,t} = 0$ yields variety growth and quality growth, respectively:¹⁶

$$n_t = \frac{(1 - \alpha)(\mu - 1) - \beta(\rho - \lambda) - [(1 - \alpha)\phi - \rho] \frac{1}{x_t l_y^*}}{\beta - \frac{\sigma}{x_t l_y^*}}, \quad (37)$$

$$z_t = \frac{\beta \left\{ \left(\alpha - \frac{\sigma}{\beta x_t l_y^*} \right) [(\mu - 1)x_t l_y^* - \phi] - [(1 - \sigma)\rho + \sigma\lambda] \right\}}{\beta - \frac{\sigma}{x_t l_y^*}}, \quad (38)$$

where we impose $r_t = r_t^e = r_t^q$. From (38), quality growth becomes positive once firm size exceeds the second threshold Ψ :

$$x_t l_y^* > \underset{\Psi}{\text{argsolve}} \left\{ [(\mu - 1)\Psi - \phi] \left(\alpha - \frac{\sigma}{\beta \Psi} \right) = (1 - \sigma)\rho + \sigma\lambda \right\}. \quad (39)$$

It implies that the economy enters the second industrial phase when $x_t > x_Z \equiv \Psi/l_y^*$.¹⁷

Combining (3), (17) and $g_t \equiv \dot{y}_t/y_t = \dot{c}_t/c_t$ yields per capita output growth in the second industrial phase as

$$g_t = \alpha [(\mu - 1)x_t l_y^* - \phi] - \rho, \quad (40)$$

which is increasing in production labor share l_y^* . Therefore, the effect of tourism expansion on transitional growth depends on how production labor share responds to an expansion in tourism, as shown in Lemma 1.

Substituting (37) into (25) yields the evolution of x_t in this phase

$$\frac{\dot{x}_t}{x_t} = \frac{1 - \sigma}{\beta - \frac{\sigma}{x_t l_y^*}} \left\{ \left[(1 - \alpha)\phi - \left(\rho + \frac{\sigma\lambda}{1 - \sigma} \right) \right] \frac{1}{x_t l_y^*} - \left[(1 - \alpha)(\mu - 1) - \beta \left(\rho + \frac{\sigma\lambda}{1 - \sigma} \right) \right] \right\}. \quad (41)$$

This shows that the law of motion of x_t is decreasing in x_t , whereas n_t is increasing in x_t since $\dot{x}_t/x_t = \lambda - (1 - \sigma)n_t$. Under the parameter condition in (24), the state variable x_t continues to grow and converges to its steady-state value

$$x^* = \frac{(1 - \alpha)\phi - \left(\rho + \frac{\sigma\lambda}{1 - \sigma} \right)}{(1 - \alpha)(\mu - 1) - \beta \left(\rho + \frac{\sigma\lambda}{1 - \sigma} \right)} \frac{1}{l_y^*}. \quad (42)$$

We substitute (42) into (40) to obtain the steady-state growth rate

$$g^* = \alpha \left[(\mu - 1) \frac{(1 - \alpha)\phi - [\rho + \sigma\lambda/(1 - \sigma)]}{(1 - \alpha)(\mu - 1) - \beta [\rho + \sigma\lambda/(1 - \sigma)]} - \phi \right] - \rho > 0, \quad (43)$$

¹⁶In Appendix B, we prove that $\beta - \sigma/(x_t l_y^*) > 0$.

¹⁷In Appendix B, we show that as the firm size increases from the first threshold to its steady-state level, the left-hand side of the equality inside the braces increases from a positive value below the right-hand side to a value above it. Therefore, the second threshold is unique.

which is independent of production labor share and therefore unaffected by tourism. This neutrality arises from the endogenous market structure in our model, under which firm size eventually converges to a stationary level. The model in Chu *et al.* (2024) is essentially a special case of the second industrial phase in our model. However, they do not consider the transition dynamics across eras/phases and simply set both the export share χ and the social return to variety σ to zero. We consider the entire transition dynamics of the economy in our model and relax these parameter restrictions to allow for a more realistic quantitative analysis. Proposition 2 summarizes the effects of tourism in the second industrial phase.

Proposition 2 (*The effect of an expansion in tourism in the second industrial phase*) *With $\delta < \xi(1 - \theta)$, leisure preference is weak; in this case, an expansion in tourism lowers the transitional growth rate. With $\delta > \xi(1 - \theta)$, leisure preference is strong; in this case, the effect of tourism expansion on transitional growth follows an inverted-U pattern: it raises (lowers) the transitional growth rate when tourism reliance is low (high). In both cases, the steady-state growth rate is unaffected by tourism.*

Proof. Equation (40) shows that the transitional growth rate in this phase is increasing in production labor share, while the steady-state growth rate in (43) is independent of it. Combining these results with Lemma 1 proves Proposition 2. ■

4 Quantitative Analysis

Our theory suggests that although tourism does not affect the long-run growth rate, it plays an important role in affecting the transition of the economy. This motivates a quantitative analysis that focuses on the transition path rather than balanced-growth outcomes. By calibrating the model to Macau’s data, we examine how tourism expansion quantitatively affects the transition dynamics in Macau. As an economy reliant on tourism, Macau has experienced significant changes in tourism share of the economy since its return to China in 1999, making it a suitable case for studying the effects of tourism on its transition dynamics.

Our model features a set of parameters: $\{\lambda, \theta, \rho, \alpha, \sigma, \mu, \beta, \chi, \delta, \phi, \xi, \tau\}$.¹⁸ We assign the population growth rate λ to 1.0%.¹⁹ The labor intensity $1 - \theta$ and the subjective discount rate ρ are, respectively, fixed at conventional values of 0.5 and 0.05. Following the calibration strategy in Iacopetta *et al.* (2019), we set the social return to variety σ and the degree of technology spillovers $1 - \alpha$ to 0.25 and 0.833, respectively. Also, we set the markup ratio μ to 1.5. In the calibration, we use data in Macau for the period 1996–2019. We calibrate the export share parameter χ using the observed export share of 6.0% of GDP.²⁰ The remaining parameters $\{\beta, \delta, \phi, \xi, \tau_{avg}\}$ are jointly calibrated by matching the following moments:²¹ 16.5%

¹⁸The preference parameter ψ only affects consumption of the imported good ι but does not affect other equilibrium outcomes.

¹⁹Data source: World Bank Data.

²⁰Data source: Fouquin and Hugot (2016), based on CEPII national trade data.

²¹We present a Jacobian analysis that reports the percentage responses of the targeted moments to a 1 percent change in each parameter in Appendix D.

for resident domestic goods consumption share of GDP in (26),²² 0.5% for the long-run TFP growth rate g^* in (43),²³ 47.5% for tourism value added as a share of total gross value added,²⁴ 49.7% for the tourism employment share $L_{s,t}/(l^*L_t) = \xi\tau$,²⁵ and 0.30 for working hours share $l^* = l_y^*/(1 - \xi\tau)$ in total available time.²⁶ Table 1 reports calibrated parameter values.

Table 1: Calibrated values of parameters

λ	θ	ρ	α	σ	μ	β	χ	δ	ϕ	τ_{avg}	ξ
0.010	0.500	0.050	0.167	0.250	1.500	1.661	0.060	9.409	1.520	0.216	2.305

The calibrated values of δ , θ and ξ imply that $\delta > \xi(1 - \theta)$. According to our model, it gives rise to an inverted-U effect of tourism expansion. Based on (29), the inverted-U pattern reaches its turning point when tourism accounts for 62.8% of gross value added.²⁷ Therefore, under our calibrated parameters, an expansion in tourism has a positive effect on Macau's economy when the tourism share is below 62.8%, but it has a negative effect when the tourism share exceeds 62.8%. This is consistent with what we observe in Macau's data: outside the period 2010–2014 (i.e., when the tourism share is below 62.8%), tourism share and TFP growth move in the same direction. However, during 2010–2014 (i.e., when the tourism share exceeds 62.8%), the relationship reverses: tourism expansion is associated with lower TFP growth.

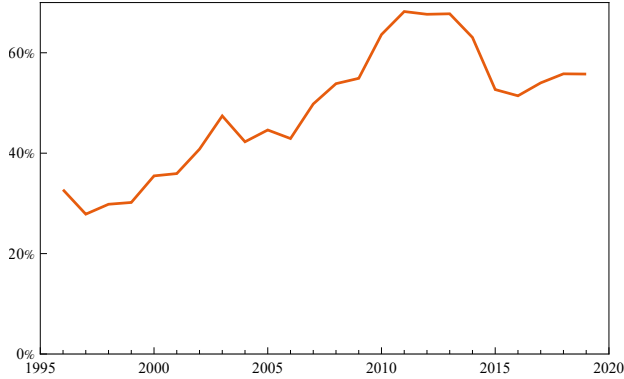


Figure 1: Tourism share of gross value added

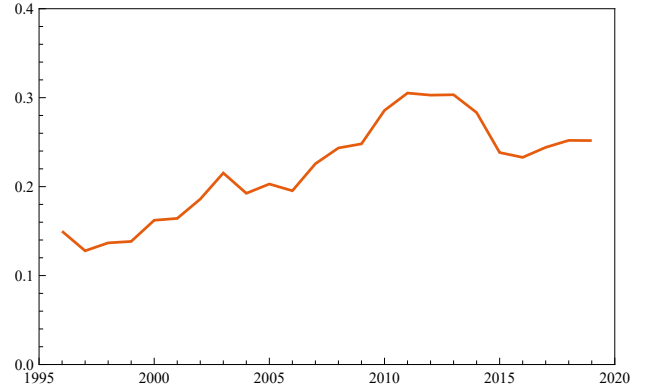


Figure 2: Calibrated path of τ

²²Macau's consumption share of GDP is much lower than in most economies because it is a tourism-dependent small open economy. In some years, Macau's exports of services account for nearly 80% of its GDP, resulting in a high net export share and thus a low consumption share of GDP. Data source: DSEC Macau Data, *Major Components of Gross Domestic Product at Current Prices*.

²³We use TFP growth instead of GDP growth in the calibration since our model emphasizes innovation-driven growth. Data source: Federal Reserve Bank of St. Louis.

²⁴In the model, the tourism share of gross value added is given by $1 - (1 - \tau)(1 - \xi\tau)/(1 - \theta\xi\tau)$. Data source: DSEC Macau Data, *Industrial Structure and Tourism Satellite Account*.

²⁵Data source: DSEC Macau Data, *Employment Survey*. It is also useful to note that DSEC identifies six tourism-related industries: gaming, retail, restaurants, hotels, transport, and travel agencies.

²⁶Since the preference parameter δ for leisure is the key parameter that determines the overall effect of tourism, we consider alternative values for working hours share in the sensitivity analysis; see Appendix D.

²⁷See Appendix D for a robustness analysis on this turning point.

In addition, we conduct a simulation to examine the quantitative effects of tourism on the transition of Macau’s economy. Figure 1 shows the tourism share in Macau’s gross value added from 1996 to 2019. The tourism share slightly decreases from 32.7% in 1996 to 30.2% in 1999. After Macau’s return to China in 1999, its tourism share rises rapidly and reaches a peak of 68.2% in 2011, and then declines with some fluctuations to 55.8% in 2019. Using the observed time series of tourism share, we take the changes in tourism share as a sequence of exogenous shocks and calibrate the path of τ (see Figure 2). In the simulation, we hold all other parameters constant and only allow the calibrated time-varying path of τ to evolve. Therefore, the simulated growth is driven solely by tourism. Under our calibrated parameters, values of x_0 between 26.96 and 27.20 are consistent with the interpretation that the economy was already in the first industrial phase in 1996 and would remain in the industrial era throughout the simulation. Here we set $x_0 = 27.00$.²⁸

Figure 3 plots the simulated growth rate from our model and the HP-filtered trend of Macau’s TFP growth. The simulated growth rate rises slowly before 1999, while the rapid expansion in tourism after Macau’s return to China drives the economy into the era of quality-driven growth from 2000 onward (i.e., the second industrial phase). Moreover, the simulated growth rate declines during 2010–2014 due to the economy’s overreliance on tourism. Our simulated growth rate aligns reasonably well with the trend of TFP growth in the data, especially before 2014.²⁹ The discrepancy after 2014 may be partly explained by Macau’s stricter standards on skilled immigration introduced in 2014 and on major investment immigration implemented in 2015, both of which likely lead to the sharp decline in TFP growth. However, it is worth noting that while the simulated TFP growth rate deviates from the data after 2014, the direction of change remains largely consistent. Our simulated TFP growth rate shows a decline from 0.033 in 2010 to 0.011 in 2019, while the data shows a decline from 0.030 to -0.069 over the same period. Under our calibrated parameters, the results suggest that tourism accounts for over one-fifth of the observed decline in Macau’s TFP growth since 2010.³⁰

We also conduct a counterfactual analysis by holding Macau’s tourism share fixed at the level observed at the time of its return to China in 1999, namely 30.2%. In this case, Macau does not enter the era of quality-driven growth until 2005. Also, its counterfactual growth rate increases very slowly and remains at a low level over time, which does not match the observed TFP growth rate in Macau. The comparison between this counterfactual experiment and the baseline simulation suggests that, after Macau’s return to China, the substantial rise in its tourism share drives the economy into the era of quality-driven growth five years earlier and results in faster technological growth.

5 Empirical Evidence

To provide empirical evidence for our theoretical predictions, in this section, we use cross-country panel data to examine whether the inverted-U relationship between tourism and economic growth also holds across economies.

²⁸We also consider other values of $x_0 \in [26.96, 27.20]$ and obtain similar dates for the transition to the era of quality-driven growth; results are available upon request.

²⁹In Appendix D, we consider alternative TFP trend measures.

³⁰See Appendix D for a robustness analysis on this result.

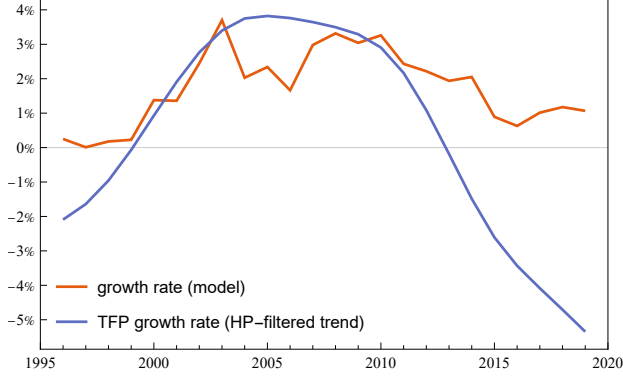


Figure 3: Simulation and data

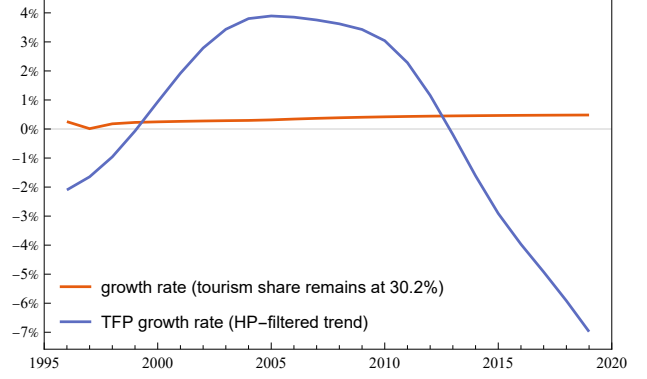


Figure 4: Counterfactual

A large empirical literature studies the link between tourism development and economic growth. Early studies by Balaguer and Cantavella-Jorda (2002) and Gunduz and Hatemi-J (2005) use time-series data to test the tourism-led growth hypothesis (TLGH) in specific countries. Subsequent work adopts cross-country panel data to examine TLGH more broadly; see for example, Brau *et al.* (2007), Lee and Chang (2008), Sequeira and Nunes (2008), Figini and Vici (2010) and Antonakakis *et al.* (2016). More recent studies document nonlinear patterns between tourism development and economic growth; see for example, Chiu and Yeh (2017), De Vita and Kyaw (2017), Zuo and Huang (2018), Antonakakis *et al.* (2019) and Dogan and Zhang (2023). However, these studies basically use GDP or GDP per capita as the growth measure, which mechanically includes tourism revenues and thus makes it difficult to examine the effect of tourism on innovation.³¹ Our empirical analysis complements this literature by employing cross-country panel data with TFP as a novel proxy for growth, which reflects technological progress more directly. This allows us to explicitly examine the relationship between tourism and innovation-driven growth. Moreover, we construct Bartik-style tourism exposure measures that plausibly captures the exogenous component of tourism-related variation. To the best of our knowledge, this is the first study to employ such a measure in this context.

Since our quantitative analysis suggests that the calibrated values of parameters satisfy $\delta > \xi(1 - \theta)$, we focus on this empirically relevant case and test the inverted-U pattern in our empirical analysis. Specifically, our estimation specification is

$$Inno_{jt} = \kappa_1 Tour_{jt} + \kappa_2 Tour_{jt}^2 + \Gamma \Phi_{jt} + \zeta_j + \zeta_t + \varepsilon_{jt}.$$

We use five-year averages data to mitigate the potential bias caused by cyclical fluctuations in annual data. $Inno_{jt}$ denotes innovation-driven growth in country j in period t , measured by the log of TFP index. Tourism development $Tour_{jt}$ is measured by the log level of tourist arrivals. Under empirically plausible parameter values, our theory expects that $\kappa_1 > 0$ and $\kappa_2 < 0$, implying an inverted-U relationship between tourism and growth. Φ_{jt} is a set of control variables, including the government expenditure share of GDP, the log level of the capital stock, the depreciation rate, and the real interest rate. We also consider country fixed effects ζ_j and period fixed effects ζ_t . ε_{jt} denotes the error term.

We merge data from Penn World Table 10.01 and World Bank Data, obtaining a sample

³¹Tzeremes (2022) employs TFP as a proxy for growth but considers only Latin American countries.

of up to 653 observations for 117 countries over 1991–2020.³² We first compute five-year averages of the original data using all available observations within each five-year period, and then transform each variable as required by the regression specification. Table 2 reports the estimation results. In all columns, κ_1 is positive and statistically significant, and κ_2 is negative and statistically significant. This empirical finding aligns with our theoretical prediction of the non-monotonic inverted-U relationship between tourism and innovation-driven growth.

Table 2: Relationship between tourism and innovation-driven growth

Dependent variable:	$Inno_{jt}$			
	(1)	(2)	(3)	(4)
$Tour_{jt}$	0.304** (0.139)	0.408*** (0.144)	0.426*** (0.147)	0.391*** (0.124)
$Tour_{jt}^2$	-0.010** (0.005)	-0.011** (0.005)	-0.013** (0.005)	-0.011** (0.004)
Control variables				✓
Country fixed effects		✓	✓	✓
Period fixed effects			✓	✓
Observations	653	653	653	653
R^2	0.1239	0.6877	0.6923	0.7379

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are reported in parentheses, clustered at the country level.

Moreover, we follow Bartik (1991), Adao *et al.* (2019) and Goldsmith-Pinkham *et al.* (2020) to construct Bartik-style tourism measures. Specifically, we interact the predetermined tourism exposure measure with the worldwide tourism shocks:

$$Tour_{jt} = T_{j,0} \times G_t,$$

where $T_{j,0}$ is measured by tourist arrivals or tourism receipts in country j before the sample period, and G_t is the growth rate of worldwide tourist arrivals. Since the tourism exposure is predetermined and worldwide tourism arrivals can be considered as aggregate shocks that are unlikely to be driven by each individual economy, this Bartik-style measure plausibly captures the exogenous component of tourism-related variation. Table 3 reports the regression results using Bartik-style tourism measures. The independent variable in columns (1)–(2) is the interaction between the predetermined tourist arrivals and aggregate tourism shocks, while the independent variable in columns (3)–(4) is the interaction between predetermined tourism receipts and aggregate tourism shocks. In all columns, κ_1 is significantly positive and κ_2 is significantly negative. Our inverted-U relationship between tourism and innovation-driven growth remains robust.

³²Summary statistics are reported in Table A.1.

Table 3: Empirical results with Bartik-style tourism exposure measures

Dependent variable:	<i>Inno_{jt}</i>			
	(1)	(2)	(3)	(4)
<i>Tour_{jt}</i>	0.361** (0.156)	0.329** (0.164)	0.330** (0.140)	0.345** (0.132)
<i>Tour_{jt}²</i>	-0.052** (0.022)	-0.046** (0.022)	-0.037** (0.017)	-0.037** (0.015)
Control variables		✓		✓
Country fixed effects	✓	✓	✓	✓
Period fixed effects	✓	✓	✓	✓
Observations	512	512	346	346
<i>R</i> ²	0.6393	0.6795	0.6428	0.7222

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are reported in parentheses, clustered at the country level.

6 Conclusion

This study develops a small-open-economy Schumpeterian growth model with a tourism sector and endogenous takeoff to study how tourism shapes the transition from stagnation to innovation-driven growth. The model highlights two opposing mechanisms through which tourism affects the economy. On the one hand, tourism expansion raises labor demand and thus total employment, which hastens takeoff and leads to faster transitional growth. On the other hand, a higher demand for tourism services reallocates labor away from production toward the tourism sector, delaying takeoff and dampening transitional growth. These two opposing effects of tourism expansion generate a non-monotonic relationship between tourism and growth. Specifically, when the economy has relatively low reliance on tourism, the employment effect dominates, which leads to positive effects on both takeoff and transitional growth. However, when the reliance on tourism is high, the reallocation effect becomes dominant, delaying takeoff and reducing transitional growth. Our quantitative analysis calibrates the model to Macau's data and suggests that tourism expansion increases Macau's TFP growth when tourism contributes less than 62.8% of the economy, but reduces its TFP growth when this turning point is exceeded. Moreover, after Macau's return to China in 1999, the rapid expansion in tourism triggers its transition to the era of quality-driven growth five years earlier. Using cross-country panel data, we provide supportive empirical evidence for this inverted-U pattern across economies: the positive association between tourism and growth weakens and eventually becomes negative as tourism reliance increases.

Overall, our findings indicate that although tourism expansion may trigger an earlier takeoff and lead to higher transitional growth, excessive dependence on tourism may delay the transition to sustained innovation-driven growth. This insight has direct policy relevance for tourism-dependent economies such as Macau, which has implemented an economic-diversification strategy aiming to reduce overdependence on tourism-related industries. Our results suggest that such a strategy may mitigate the adverse growth effects of excessive tourism dependence, although its welfare implications remain beyond the scope of our analysis.

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Appendix

A Data

Table A.1: Summary statistics

Variable	Observations	Mean	Sd	Min	Max
Log TFP index	653	-0.038	0.199	-1.493	0.730
Log tourist arrivals	653	14.837	1.906	8.200	19.122
Log capital stock	653	12.958	2.008	8.403	18.294
Government expenditure share of GDP	653	0.178	0.067	0.007	0.616
Capital depreciation rate	653	0.043	0.011	0.021	0.100
Real interest rate	653	0.116	0.076	0.010	0.624

Data source: World Bank Data for tourist arrivals. Penn World Table for other variables.

B Proofs

Proof of Lemma 1. In the pre-industrial era, innovation is absent and the asset value is zero, so there is no asset accumulation. We combine $\dot{a}_t = 0$, (2), (8), (21) to obtain

$$\frac{c_t}{y_t} = \frac{w_t l_t - p_t l_t}{y_t} = 1 - \theta - \tau - \chi, \quad (\text{B.1})$$

which is positive if and only if $1 - \theta - \tau - \chi > 0$.

In the industrial era, the free-entry condition (18) holds due to the entry of new firms. Imposing symmetry, we use (9), (13), $N_t V_t = a_t L_t$ and the free-entry condition to derive

$$a_t = \frac{\beta\theta}{\mu} y_t. \quad (\text{B.2})$$

Combining (2), (3), (8), (21) and (B.2) yields

$$\frac{\dot{c}_t}{c_t} - \frac{\dot{y}_t}{y_t} = \frac{\mu}{\beta\theta} \frac{c_t}{y_t} - \left[\frac{\mu(1 - \theta - \tau - \chi)}{\beta\theta} + \rho - \lambda \right], \quad (\text{B.3})$$

which says that the consumption-output ratio jumps to its steady-state value $(c_t/y_t)^*$ and remains constant in the industrial era. We then use the supply of labor (4), the demand for labor (8) and $l_{y,t} = (1 - \xi\tau)l_t$ to derive the expression for production labor share as

$$l_{y,t} = \left(\frac{1}{1 - \xi\tau} + \frac{\delta}{1 - \theta} \frac{c_t}{y_t} \right)^{-1}. \quad (\text{B.4})$$

Substituting the stationary values of the consumption-output ratio into (B.4) yields the stationary values of production labor share for both the pre-industrial and industrial eras, as summarized in Lemma 1. ■

Verification of the transversality condition. The current-value Hamiltonian of the representative household is

$$\mathcal{H}_t = \ln c_t + \delta \ln(1 - l_t) + \psi \frac{(l_t)^{1-\epsilon}}{1-\epsilon} + \varphi_t [(r_t - \lambda)a_t + w_t l_t - c_t - p_t l_t], \quad (\text{B.5})$$

where φ_t is the current-value multiplier. The first-order condition with respect to consumption is

$$\frac{\partial \mathcal{H}_t}{\partial c_t} = 0 \Rightarrow \frac{1}{c_t} = \varphi_t. \quad (\text{B.6})$$

The transversality condition is

$$\lim_{t \rightarrow \infty} e^{-(\rho-\lambda)t} \varphi_t a_t = 0. \quad (\text{B.7})$$

Using (B.6), the left-hand side of (B.7) can be written as

$$\lim_{t \rightarrow \infty} e^{-(\rho-\lambda)t} \frac{a_t}{c_t}. \quad (\text{B.8})$$

According to Lemma 1, when $t \rightarrow \infty$, the consumption-output ratio stays at its steady-state value $(\rho - \lambda)\beta\theta/\mu + 1 - \theta - \tau - \chi$. Using (B.2), we derive (B.7) as

$$\lim_{t \rightarrow \infty} e^{-(\rho-\lambda)t} \frac{1}{\rho - \lambda + \frac{\mu}{\beta\theta}(1 - \theta - \tau - \chi)} = 0. \quad (\text{B.9})$$

Since $\lambda \in (0, \rho)$, we have $e^{-(\rho-\lambda)t} \rightarrow 0$. This verifies the transversality condition. ■

Positivity of the denominator $\beta - \sigma/(x_t l_y^*)$. According to (33), when the economy is in the industrial era, firm size $x_t l_y^*$ is greater than the first threshold $\phi/[\mu - 1 - \beta(\rho - \lambda)]$. Therefore, we have

$$x_t l_y^* > \frac{\phi}{\mu - 1 - \beta(\rho - \lambda)} > \frac{\phi}{\mu - 1}. \quad (\text{B.10})$$

Given $\beta\phi > \mu - 1$ in (24) and $\sigma \in (0, 1)$, we obtain

$$\frac{\phi}{\mu - 1} > \frac{1}{\beta} > \frac{\sigma}{\beta}. \quad (\text{B.11})$$

Combining (B.10) and (B.11) yields

$$x_t l_y^* > \frac{\sigma}{\beta} \Leftrightarrow \beta - \frac{\sigma}{x_t l_y^*} > 0. \quad (\text{B.12})$$

■

Phase ordering. In the main text, our model considers a more realistic sequence in which variety expansion happens before quality improvement. In this appendix, we consider the alternative sequence and then derive the parameter condition that ensures the more realistic sequence.

In the case where quality improvement happens before variety expansion, the free-entry condition does not hold when $n_t = 0$ and $z_t > 0$. Therefore, the consumption-output ratio is

not constant in this phase. Using (2), (10), (13), (15), (16), (19), $N_t V_t = a_t L_t$ and imposing symmetry yield

$$\frac{c_t}{y_t} = 1 - \theta - \tau - \chi + \frac{\theta}{\mu} \left(\mu - 1 - \frac{\phi + z_t}{x_t l_{y,t}} \right). \quad (\text{B.13})$$

We use (3), (17), (23) and $n_t = 0$ to obtain the transitional growth rate

$$g_t \equiv \frac{\dot{y}_t}{y_t} = \frac{\dot{c}_t}{c_t} - \left(\frac{\dot{c}_t}{c_t} - \frac{\dot{y}_t}{y_t} \right) = \alpha[(\mu - 1)x_t l_{y,t} - \phi] - \rho - \left(\frac{\dot{c}_t}{c_t} - \frac{\dot{y}_t}{y_t} \right) = \frac{\dot{l}_{y,t}}{l_{y,t}} + z_t. \quad (\text{B.14})$$

It is useful to note that $z_t = 0$ when the firm size is at the threshold for quality improvement. In other words, the firm size $x_t l_{y,t}$ satisfies the condition $\alpha[(\mu - 1)x_t l_{y,t} - \phi] = \rho + \dot{c}_t/c_t - \dot{y}_t/y_t + \dot{l}_{y,t}/l_{y,t}$ at the transition from stagnation to quality-improving innovation. Substituting (B.4) and $z_t = 0$ into (B.13) yields

$$\frac{c_t}{y_t} = \frac{1 - \theta - \tau - \chi + \frac{\theta}{\mu}(\mu - 1) - \frac{\theta}{\mu} \frac{1}{1 - \xi \tau} \frac{\phi}{x_t}}{1 + \frac{\theta}{\mu} \frac{\delta}{1 - \theta} \frac{\phi}{x_t}}. \quad (\text{B.15})$$

We substitute (B.15) into (B.4) to derive the production labor share

$$l_{y,t} = \frac{1 + \frac{\theta}{\mu} \frac{\delta}{1 - \theta} \frac{\phi}{x_t}}{\frac{1}{1 - \xi \tau} + \frac{\delta}{1 - \theta} \left[1 - \theta - \tau - \chi + \frac{\theta}{\mu}(\mu - 1) \right]}. \quad (\text{B.16})$$

Since $\dot{x}_t/x_t = \lambda$, we take the log derivative of (B.15) with respect to time to derive

$$\frac{\dot{c}_t}{c_t} - \frac{\dot{y}_t}{y_t} = \frac{\frac{\theta}{\mu} \frac{1}{1 - \xi \tau} \frac{\phi}{x_t} \lambda}{1 - \theta - \tau - \chi + \frac{\theta}{\mu}(\mu - 1) - \frac{\theta}{\mu} \frac{1}{1 - \xi \tau} \frac{\phi}{x_t}} + \frac{\frac{\theta}{\mu} \frac{\delta}{1 - \theta} \frac{\phi}{x_t} \lambda}{1 + \frac{\theta}{\mu} \frac{\delta}{1 - \theta} \frac{\phi}{x_t}}, \quad (\text{B.17})$$

which is decreasing in x_t . Taking log derivative of (B.16) yields

$$\frac{\dot{l}_{y,t}}{l_{y,t}} = - \frac{\frac{\theta}{\mu} \frac{\delta}{1 - \theta} \frac{\phi}{x_t} \lambda}{1 + \frac{\theta}{\mu} \frac{\delta}{1 - \theta} \frac{\phi}{x_t}}. \quad (\text{B.18})$$

In the main text, according to (33), the threshold for variety expansion is

$$x_N l_y^* = \frac{\phi}{\mu - 1 - \beta(\rho - \lambda)}, \quad (\text{B.19})$$

where l_y^* is given by the second line of (27). Variety expansion happens before quality improvement implies that, when the firm size reaches the threshold $\phi/(\mu - 1 - \beta(\rho - \lambda))$ (i.e., when $x_t = x_N$) that triggers variety expansion, it is not sufficiently large to trigger quality improvement. Therefore, the following inequality must hold

$$\alpha[(\mu - 1)x_N l_y^* - \phi] < \rho + \left(\frac{\dot{c}_t}{c_t} - \frac{\dot{y}_t}{y_t} \right) + \frac{\dot{l}_{y,t}}{l_{y,t}}. \quad (\text{B.20})$$

Substituting (B.17), (B.18) and (B.19) into (B.20) yields the following parameter restriction:

$$\alpha < \bar{\alpha} \equiv \frac{\mu - 1 - \beta(\rho - \lambda)}{\beta(\rho - \lambda)\phi} \left\{ \rho + \frac{\frac{\theta}{\mu} \frac{1}{1 - \xi \tau} [\mu - 1 - \beta(\rho - \lambda)] \lambda}{\left[1 - \theta - \tau - \chi + \frac{\theta}{\mu} \beta(\rho - \lambda) \right] \left\{ \frac{1}{1 - \xi \tau} + \frac{\delta}{1 - \theta} \left[1 - \theta - \tau - \chi + \frac{\theta}{\mu}(\mu - 1) \right] \right\}} \right\}. \quad (\text{B.21})$$

■

Proof of $x_N < x_Z < x^*$ in our model. Using (39), if $x_N < x_Z$, we have

$$[(\mu - 1)x_N l_y^* - \phi] \left(\alpha - \frac{\sigma}{\beta x_N l_y^*} \right) < (1 - \sigma)\rho + \sigma\lambda, \quad (\text{B.22})$$

which implies that the firm size $x_N l_y^*$ is too small to activate quality improvement. Using (B.19), we rearrange (B.22) as

$$\alpha < \frac{\mu - 1 - \beta(\rho - \lambda)}{\phi\beta(\rho - \lambda)}\rho, \quad (\text{B.23})$$

which holds under the parameter condition given in (B.21).

In addition, if $x^* > x_Z$, we have

$$[(\mu - 1)x^* l_y^* - \phi] \left(\alpha - \frac{\sigma}{\beta x^* l_y^*} \right) > (1 - \sigma)\rho + \sigma\lambda, \quad (\text{B.24})$$

which implies that quality improvement is active when the firm size is $x^* l_y^*$. We use (42) and rearrange (B.24) as

$$\beta \left[(1 - \alpha)\phi - \left(\rho + \frac{\sigma\lambda}{1 - \sigma} \right) \right] - \sigma \left[(1 - \alpha)(\mu - 1) - \beta \left(\rho + \frac{\sigma\lambda}{1 - \sigma} \right) \right] > 0 \quad (\text{B.25})$$

which must hold under the parameter condition (24). Given that the left-hand side of the equality inside the braces of (39) is increasing in firm size, the above derivations imply that the second threshold $x_Z l_y^*$ is unique. ■

C An Alternative Scenario in the Pre-industrial Era

We further explore an alternative scenario in our model, where monopolistic firms operate even in the pre-industrial era. Using (10), (13), $X_t/Z_t = x_t l_{y,t}$, and imposing symmetry, we express the profit of monopolistic firms as

$$\Pi_t = [(\mu - 1)x_t l_{y,t} - \phi] Z_t, \quad (\text{C.1})$$

which is increasing in firm size $x_t l_{y,t}$. In this scenario, monopolistic firms operate and earn positive profits in the pre-industrial era. Specifically, $\Pi_t > 0$ when $x_0 \leq x_t < x_N$. Based on (C.1), this condition is satisfied if and only if

$$x_0 l_{y,0} > \frac{\phi}{\mu - 1}, \quad (\text{C.2})$$

where $x_0 l_{y,0}$ is the initial firm size.

Since firms do not invest in innovation (i.e., $R_t = 0$) in the pre-industrial era, (19) becomes

$$r_t = \frac{\Pi_t}{V_t} + \frac{\dot{V}_t}{V_t}. \quad (\text{C.3})$$

The aggregate value of firms is equal to the total value of household assets, i.e., $a_t L_t = N_t V_t$. We then combine (2), $a_t L_t = N_t V_t$ and $n_t = 0$ to derive

$$\frac{\dot{a}_t}{a_t} = \frac{\dot{V}_t}{V_t} - \lambda = r_t - \lambda + \frac{w_t l_t - c_t - p_t \iota_t}{a_t}. \quad (\text{C.4})$$

Using (C.3) and (C.4) yields

$$c_t = \frac{\Pi_t}{V_t} a_t + w_t l_t - p_t \iota_t = \frac{N_t}{L_t} \Pi_t + (1 - \theta - \tau - \chi) y_t, \quad (\text{C.5})$$

where the second equality uses $a_t L_t = N_t V_t$, $p_t \iota_t = \tau y_t + w_t l_{s,t} + \chi y_t$, $w_t l_{y,t} = (1 - \theta) y_t$ and $l_{y,t} + l_{s,t} = l_t$. We then impose symmetry, and substitute (10) and (13) into (C.5) to derive

$$c_t = \frac{N_t}{L_t} [(\mu - 1)X_t - \phi Z_t] + (1 - \theta - \tau - \chi) y_t = \frac{\theta}{\mu} \left(\mu - 1 - \frac{\phi}{x_t l_{y,t}} \right) y_t + (1 - \theta - \tau - \chi) y_t, \quad (\text{C.6})$$

where the second equality uses $\theta Y_t = \mu N_t X_t$ and $X_t/Z_t = x_t l_{y,t}$. We rearrange (C.6) to express the consumption-output ratio in the pre-industrial era as

$$\frac{c_t}{y_t} = \frac{\theta}{\mu} \left(\mu - 1 - \frac{\phi}{x_t l_{y,t}} \right) + 1 - \theta - \tau - \chi. \quad (\text{C.7})$$

The consumption-output ratio would increase from $1 - \theta - \tau - \chi$ to $1 - \theta - \tau - \chi + (\rho - \lambda)\beta\theta/\mu$ if firm size $x_t l_{y,t}$ starts from $\phi/(\mu - 1)$ and increases toward $\phi/[\mu - 1 - \beta(\rho - \lambda)]$, which is the threshold for variety expansion as shown in (33). Finally, we use (B.4) and (C.7) to derive the expression for firm size $x_t l_{y,t}$ in the pre-industrial era

$$x_t l_{y,t} = \frac{x_t + \frac{\theta}{1-\theta} \frac{\delta\phi}{\mu}}{\frac{1}{1-\xi\tau} + \delta \left(1 + \frac{\mu-1}{\mu} \frac{\theta}{1-\theta} - \frac{\tau+\chi}{1-\theta} \right)}. \quad (\text{C.8})$$

Given the law of motion for x_t in (25) and $n_t = 0$ in the pre-industrial era, firm size $x_t l_{y,t}$ gradually increases toward the first threshold $\phi/[\mu - 1 - \beta(\rho - \lambda)]$ and then triggers takeoff as in our baseline model. The only difference here is that, as x_t increases over time, $l_{y,t}$ in (C.8) gradually decreases (instead of jumping at the time of the takeoff). The additional dynamics of $l_{y,t}$ imply negative growth in per capita final output in the pre-industrial era, which is less realistic than the dynamics in the baseline model.

D Sensitivity Analysis

In this appendix, we conduct sensitivity analysis of our quantitative results. First, we consider alternative TFP growth trend measures. Second, since the preference parameter δ for leisure plays a key role in determining the overall effect of tourism, we consider alternative values for working hours share as the target moment, namely, 0.27 and 0.33. We then recalibrate and resimulate the model to examine how the quantitative results change. Finally, we report a Jacobian analysis to assess the sensitivity of the target moments to changes in the calibrated parameters.

D.1 Alternative TFP Growth Measures

In addition to HP-filtered TFP growth, we consider LOESS TFP growth in Figure D.1 and CF-filtered TFP growth in Figure D.2. LOESS TFP growth declines from 0.045 in 2010 to -0.059 in 2019, and CF-filtered TFP growth declines from 0.045 in 2010 to -0.063 in 2019. Recall that our simulated growth rate shows a decline from 0.033 to 0.011 over the same period. The decline in our simulated growth rate is about one-fifth of the decline in these two alternative TFP growth measures.

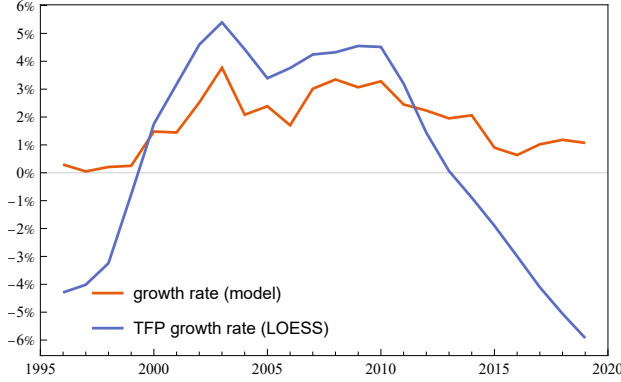


Figure D.1: Simulation and data (LOESS)

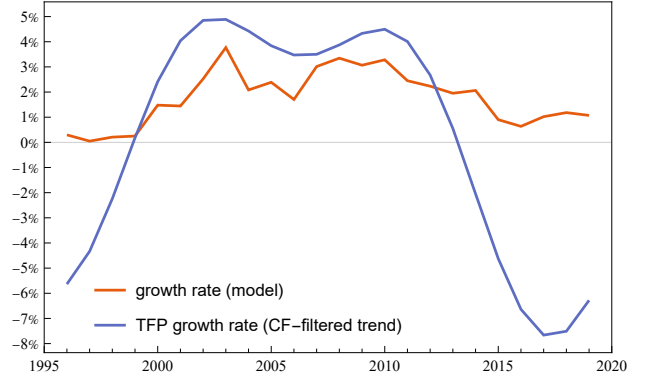


Figure D.2: Simulation and CF-filtered trend

D.2 Alternative Target Moments

Since the target moment of working hours share in total available time is closely related to the leisure preference parameter δ , which determines the overall effect of tourism, we consider alternative values for working hours share to obtain different calibrated values of δ and examine whether our quantitative results change substantially.

We first consider a working hours share l^* of 0.27, which is 10% lower than the target moment used in Section 4. Holding all other target moments fixed and changing only the working-hours share does not affect the calibrated values of other parameters but changes the calibrated value of δ to 10.902. Using this set of calibrated parameters, the inverted-U pattern reaches its turning point when tourism accounts for 65.3% of gross value added. In the simulation, the economy enters the era of quality-driven growth in 2000, and the simulated growth rate declines from 0.041 in 2010 to 0.008 in 2019, accounting for about 30 percent of the decline in TFP growth observed in the data; see Figure D.3.

We then consider a working hours share l^* of 0.33, which is 10% greater than the target moment used in Section 4. Similarly, this leaves the calibrated values of other parameters unchanged but changes the calibrated value of δ to 8.187. Using this set of calibrated parameters, the inverted-U pattern reaches its turning point when tourism accounts for 60.2% of gross value added. In the simulation, the economy also enters the era of quality-driven growth in 2000, and the simulated growth rate declines from 0.024 in 2010 to 0.011 in 2019, accounting for about 15 percent of the decline in TFP growth observed in the data; see Figure D.4.

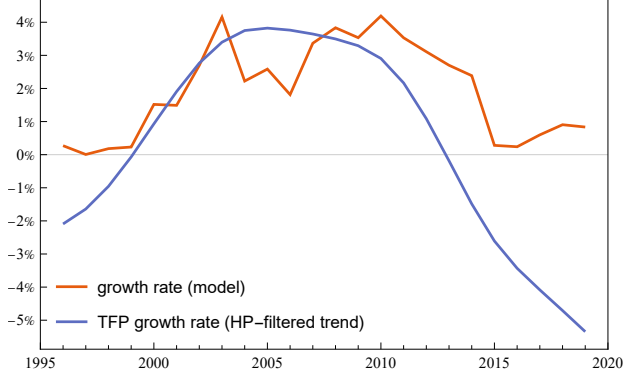


Figure D.3: Simulation ($\delta = 10.902$)

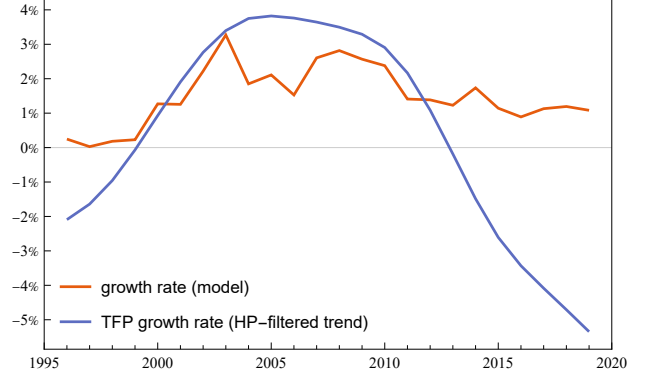


Figure D.4: Simulation ($\delta = 8.187$)

D.3 Jacobian Analysis

To clarify the mapping between the calibrated parameters and the target moments, Table D.1 summarizes the expressions for the moments. To illustrate the local identification of the calibration, we conduct a Jacobian analysis, which reports how each target moment responds to changes in each parameter around the baseline calibration; see Table D.2.

Table D.1: Expressions for target moments

Moment	Expression
resident domestic goods consumption / GDP	$\frac{1-\xi\tau}{1-\theta\xi\tau} \left[\frac{(\rho-\lambda)\beta\theta}{\mu} + 1 - \theta - \tau - \chi \right]$
working hours share in available time	$\left\{ 1 + (1 - \xi\tau) \frac{\delta}{1-\theta} \left[\frac{(\rho-\lambda)\beta\theta}{\mu} + 1 - \theta - \tau - \chi \right] \right\}^{-1}$
tourism employment share	$\xi\tau$
tourism value added share of gross value added	$1 - \frac{(1-\tau)(1-\xi\tau)}{1-\theta\xi\tau}$
long-run TFP growth rate	$\alpha \left[(\mu - 1) \frac{(1-\alpha)\phi - [\rho + \sigma\lambda/(1-\sigma)]}{(1-\alpha)(\mu-1) - \beta[\rho + \sigma\lambda/(1-\sigma)]} - \phi \right] - \rho$

Notes: Given the assigned values of $\{\lambda, \theta, \rho, \sigma, \alpha, \mu, \chi\}$, we jointly calibrate $\{\beta, \delta, \phi, \xi, \tau_{avg}\}$ by matching the target moments.

Table D.2: Percentage responses of moments to one-percent changes in parameters

Moment	β	δ	ϕ	ξ	τ
resident domestic goods consumption / GDP	0.09	0.00	0.00	-0.66	-1.53
working hours share in available time	-0.06	-0.70	0.00	0.70	1.31
tourism employment share	0.00	0.00	0.00	1.00	1.00
tourism value added share of gross value added	0.00	0.00	0.00	0.73	1.03
long-run TFP growth rate	16.73	0.00	13.72	0.00	0.00

Notes: The numbers in the table denote the percentage change in each moment that results from a one-percent change in each parameter. The Jacobian is evaluated at the baseline calibration, with $\tau = \tau_{avg}$.